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Abdillah Muharrarul

Data Scientist

Portfolio: [nekyollas-portfolio](#)
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An Informatics Engineering student at Universitas Airlangga with a strong interest in Data Science, Machine Learning, and Artificial Intelligence, supported by a solid foundation in backend development. Passionate about exploring new technologies and turning knowledge into AI-driven solutions that create tangible, measurable impact for society.

EDUCATION

Bachelor of Applied Science (D4) in Informatics Engineering, Universitas Airlangga 2023 — Present
GPA: 3.31/4.00 | Focusing on Data Science, Machine Learning, and Artificial Intelligence.

High School Diploma (Science/MIPA), SMAN 1 Wirosari 2019 — 2022
Average Score: 82.3/100

SKILLS

Languages	Go, Python, TypeScript, JavaScript
Frameworks	Next.js, React Native, Node.js, Express.js, TailwindCSS, Streamlit
AI / ML	Numpy, Pandas, Scikit-learn, TensorFlow, PyTorch, OpenCV, HuggingFace Transformers
Databases	MySQL, PostgreSQL, MongoDB, Supabase
Tools & Platforms	Git, Vercel, Postman, Swagger, Figma, $\LaTeX$

TECHNICAL EXPERIENCE

Backend Developer — Student Achievement Credit System (SKP) 2025
Universitas Airlangga Surabaya, Indonesia

- Built a REST API backend for reporting and verifying student achievements with role-based access control using Go and Fiber.
- Designed a dual-database architecture: PostgreSQL for RBAC and MongoDB for dynamic achievement data.
- Implemented fine-grained RBAC with 3 roles (Admin, Student, Advisor) and 6+ permissions using JWT middleware.
- Built achievement workflow lifecycle: draft → submitted → verified/rejected with state transition notifications.
- Delivered full Swagger API documentation covering 20+ endpoints.
- Wrote unit tests for all service functions with mocked external dependencies to ensure system reliability.

Backend Developer — Bengkel Brainrot (Campus Workshop Directory) 2026
Cloud Computing Final Project, Universitas Airlangga Surabaya, Indonesia

- Developed a cloud-native mobile directory app for finding nearby workshops with real-time GPS and turn-by-turn navigation.
- Built a serverless Node.js + Express API on Vercel as a security gatekeeper, preventing direct Supabase access from client.
- Integrated OSRM for turn-by-turn driving directions in Bahasa Indonesia with 30-meter arrival detection.
- Seeded 66 workshop locations from CSV, displayed as live GPS markers on Google Maps.
- Applied OWASP Top 10 API Security mitigations with strict input validation and centralized error handling.

Backend Developer — CloudTrack (IoT Telemetry Platform) 2026
Cloud Computing Midterm Project, Universitas Airlangga Surabaya, Indonesia

- Built a cloud-based IoT platform integrating QR attendance, accelerometer telemetry, and GPS tracking into a unified API.
- Implemented a stateless token-based QR attendance system with expiry validation for tamper-proof check-in.
- Developed batch accelerometer ingestion pipeline supporting multiple samples per request.
- Rendered GPS tracking history as polylines on an interactive map with latest-position markers.
- Ensured cross-group API compatibility following API Contract Simple v1 standard for swap testing.

Data Analyst — Apriori Market Basket Analysis 2026
Universitas Airlangga Surabaya, Indonesia

- Built a Streamlit web app implementing the Apriori algorithm for association rule mining on retail transaction data.
- Developed full ARM pipeline: CSV parsing → one-hot encoding → frequent itemset mining → rule generation.
- Implemented interactive Min Support & Min Confidence parameter tuning via sidebar sliders.
- Created automated business insight section with cross-selling and product placement recommendations.

Developer — Nekylla Portfolio 2026
Personal Project Surabaya, Indonesia

- Built a modern, data-driven portfolio with Next.js 16, TypeScript, TailwindCSS, and Framer Motion.
- Implemented dynamic SSG project detail pages via dynamic routing with generateStaticParams.
- Created LinkedIn-style certificate modal with embedded PDF preview and download functionality.
- Designed fully responsive, mobile-first layout with SEO-optimized static pages.

EXPERIENCE

- Student Executive Board, Universitas Airlangga, Creative Product & Investment Officer 2024 — 2025
- Co-organized the 'Airlangga Business Forum', a university-wide business plan competition aimed at fostering innovation and student entrepreneurship.
 - Successfully held an investment and cryptocurrency seminar in collaboration with industry partners like Coinkami to improve financial literacy among students.
 - Facilitated networking opportunities between student innovators and external industrial sectors to boost the economic value of local creative products.
- Student Executive Board, Faculty of Vocational Studies, UNAIR, Research & Innovation Officer 2024 — 2025
- Organized the 'Vokasi Goes to OLIVIA 2024' socialization program to increase student awareness and participation in the Indonesian Vocational Olympiad.
 - Managed a data tracking system using digital forms to identify, record, and monitor outstanding students' achievements monthly.
 - Curated and disseminated information about academic and scientific competitions to support student development.
 - Collaborated with the Publications and Information Division to design and publish monthly appreciation posters.
- UNAIR Esports, Esports Athlete – Goldlaner (Mobile Legends: Bang Bang) 2024 — 2025
- 1st Place – Rector's Cup UPN Veteran Jawa Timur x AMN Cup 2023
 - 1st Place – National Mogmain Festival Esports Tournament 2024
 - 1st Place – Airlangga Esports League 2025
 - 1st Place – UNUSA Cup Series 2024
 - 1st Place – UNUSA Esports Future Legend 2025
 - 1st Place – Mechanical Engineering E-Sport Competition UNESA 2024
 - 1st Place – Honda Esports Competition 2024
 - 2nd Place – Honda Esports Competition 2025
 - 2nd Place – Management Esports Competition UNAIR 2024
 - 3rd Place – Airlangga Esports Competition 2024
 - 3rd Place – Epiclair Esports Tournament 2025
- Student Council, SMAN 1 Wirosari, Environmental & Academic Division Officer 2020 — 2021
- Initiated a green campaign by upcycling used bottles into plant pots to enhance school eco-friendliness.
 - Spearheaded a 'Bring Your Own Tumbler' campaign, successfully reducing single-use plastic waste on campus.
 - Curated and distributed a database of past midterm and final exam papers to support student test preparation.